

Structural Sparsity for Quantum Simulation: A Synthesis of the GeoVac Quantum-Computing Arc*

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The Geometric Vacuum (GeoVac) framework discretizes electronic structure on natural geometries, producing sparse graph Hamiltonians whose angular content is fixed by Gaunt selection rules. This synthesis assembles the framework’s quantum-computing arc—the four Group 4 papers—into a single resource-oriented narrative. The spine is: the qubit encoding and its Pauli/1-norm scaling [2]; the benchmarked resource library across three periodic-table rows [4]; the S_N representation-theoretic foundation that classifies the atomic building blocks [3]; and the nuclear shell-model extension that delineates exactly which part of the framework transfers [6]. The unifying value proposition is *structural*, not accuracy-based: after the Jordan–Wigner transformation the GeoVac lattice Hamiltonian yields $O(Q^{3.15})$ Pauli terms for atoms and $O(Q^{2.5})$ for composed multi-center molecules (versus $O(Q^{3.9-4.3})$ for Gaussian baselines), with a sub-quadratic Pauli 1-norm $\lambda = O(Q^{1.69})$ ($R^2 = 0.997$) that controls Trotter depth and qubitization query cost. Two disciplines run through the arc. *First*, we separate the robust *scaling* exponents—unchanged under the choice of electron-repulsion selection rule—from the *absolute multipliers*, which depend on a stated pair-diagonal angular approximation and are disclosed as such. *Second*, every comparison is made at *matched qubit count*, not matched accuracy: the composed basis carries classical equilibrium-geometry errors of 5–26%, so the framework is positioned for fixed-geometry NISQ/VQE simulation (Pauli count, QWC measurement groups), not for fault-tolerant geometry optimisation. The nuclear extension is honest in the same way: its binding energies are encoding-validation benchmarks far from experiment, and the Fock projection rigidity theorem states precisely that the conformal S^3 structure does *not* transfer to the harmonic-oscillator basis—only the angular machinery and the qubit encoding do.

I. INTRODUCTION: A STRUCTURAL VALUE PROPOSITION

The classical-solver investigation that occupies the GeoVac quantum-chemistry arc (Group 2) reached a clear verdict: the natural-geometry hierarchy characterises what discrete graph Hamiltonians can and cannot achieve for ground-state potential-energy surfaces, and on the accuracy axis it is not competitive with production quantum chemistry. The framework’s computational value lies elsewhere. The same construction that makes the classical accuracy bounded—a single-center angular basis with Gaunt-enforced selection rules and a pseudopotential-partitioned composed geometry—makes the resulting *qubit* Hamiltonians structurally sparse. This synthesis collects the four papers that develop and benchmark that observation.

The reader should hold two framings in mind throughout. First, the advantage is in *resource scaling and term counts*, measured at *matched qubit count*, not in energy accuracy: at $n_{\max} = 2$ the composed basis carries 5–26% classical equilibrium-geometry errors (LiH/BeH₂/H₂O), so the natural use is single-point simulation at a fixed nuclear geometry, where the sparsity is free and the geometry error does not enter. Second, the framework targets the NISQ/VQE regime—where Pauli term count

and the number of qubitwise-commuting (QWC) measurement groups dominate cost—rather than the fault-tolerant QPE regime, although the sub-quadratic Pauli 1-norm is also the relevant quantity for qubitization.

The four papers form a dependency spine. Paper 14 [2] establishes the encoding and its scaling laws. Paper 20 [4] benchmarks a 37-system library (35 composed molecules + He + H₂) across three periodic-table rows and exports them to the standard quantum-software ecosystems. Paper 16 [3] supplies the S_N representation-theoretic classification of the atomic building blocks that the composed builder assembles. Paper 23 [6] extends the angular machinery to the nuclear shell model and proves, via the Fock rigidity theorem, the boundary of what transfers.

II. THE ANGULAR-SPARSITY MECHANISM

The structural origin of every resource result in this arc is a single fact about the two-electron integral (ERI) tensor. In a single-center basis indexed by (n, ℓ, m) , the angular part of each ERI is a product of Gaunt coefficients, and a Gaunt coefficient vanishes unless its three angular momenta satisfy the $3j$ triangle inequality and the magnetic selection rule. As the number of spatial orbitals M grows, the fraction of nonzero ERIs therefore *decays*—empirically as $\sim 1/M$ ($M^{-0.85}$)—whereas a Gaussian ERI tensor remains near-dense at $O(1)$ occupancy. This is the potential-independent angular-sparsity theorem of Paper 22 (Group 3): the ERI density depends only

* Group 4 synthesis in the Geometric Vacuum series

on $\ell_{\max}$, not on the radial potential $V(r)$. The sparsity survives the Jordan–Wigner transformation because it is carried by the angular labels, which JW does not mix.

a. The selection rule actually realised, and its disclosure. A precise statement of the sparsity requires naming *which* angular selection rule the production composed pipeline realises. The exact Coulomb rule is the global- M_L rule, $m_a + m_b = m_c + m_d$. The production composed builder realises instead the *pair-diagonal* rule $m_a = m_c \wedge m_b = m_d$: a legitimate sparsifying approximation that retains the diagonal magnetic channels and drops the genuinely nonzero “m-swap” integrals (e.g. $\langle p_{+1}p_{-1}|p_{-1}p_{+1} \rangle$, angular factor ≈ 0.49). This distinction matters for the *magnitude* of the sparsity advantage but not for its *scaling*. A library-wide build-only sweep shows that switching to the exact global- M_L rule multiplies the composed Pauli count by a *constant factor within each valence class*— $2.51\times$ for main-group (s, p) molecules and $3.25\times$ for d -block transition-metal hydrides—because the dropped m-swaps are a property of the valence angular content, not of the molecular geometry. Consequently:

- the *scaling* exponents and the exact linearity $N_{\text{Pauli}} = cQ$ are unchanged (a constant prefactor leaves the log-log slope invariant);
- the *absolute multipliers* and any absolute market-test line re-price by the constant factor, and are reported below with the rule they assume stated explicitly.

We adopt the convention of *disclosing* the pair-diagonal rule and reporting both the pair-diagonal multiplier and its global- M_L re-pricing, rather than presenting the pair-diagonal count as exact angular-selection-rule sparsity.

III. THE ENCODING AND ITS SCALING LAWS (PAPER 14)

Paper 14 transforms the GeoVac lattice Hamiltonian to qubits via Jordan–Wigner and characterises the resulting Pauli decomposition by several operationally distinct metrics.

a. Term count. For single-center atoms the Pauli term count scales as $O(Q^{3.15})$ in the qubit number Q , compared with Gaussian molecular Pauli-scaling exponents in the $Q^{3.9}$ – $Q^{4.3}$ range; the specific electronic-structure values $Q^{4.25}$ (LiH) and $Q^{3.92}$ (H_2O), and the range they bracket, are GeoVac’s own log-log fit of the published Gaussian Pauli counts of Trenev et al. [7] (Table 5, Appendix B; Jordan–Wigner with 2-qubit reduction). For composed multi-center molecules the pseudopotential-partitioned architecture produces block-diagonal ERIs and a *universal* $O(Q^{2.5})$ scaling, with an exponent spread of only 0.02 across the benchmarked molecules—independent of molecular topology. Across the library the composed count is, remarkably, *exactly*

linear in the qubit count,

$$N_{\text{Pauli}} = cQ, \quad c = 11.10 \text{ (main group)}, \quad 9.23 \text{ (} d\text{-block)}, \quad (1)$$

under the production pair-diagonal rule. (Under the exact global- M_L rule the coefficients become 27.9 and 30.0; note that the ordering reverses—the d -block, which is *sparser* under the pair-diagonal rule, is in fact *denser* under the exact rule, because higher angular momentum carries more m-swap channels to drop. The pair-diagonal d -block sparsity is thus an artifact of the approximation, not a selection-rule property, and is labeled as such.)

b. Measurement and Trotter cost. Two further metrics govern practical cost. The number of QWC measurement groups—the dominant factor in the measurement overhead of variational algorithms—scales as $O(Q^{3.36})$. The Pauli 1-norm $\lambda = \sum_i |c_i|$, which bounds both Trotter step counts and qubitization query complexity, scales as $O(Q^{1.69})$ with $R^2 = 0.997$. This sub-quadratic 1-norm scaling is the key fault-tolerant-relevant result: Trotter circuit depth grows far more slowly with system size than the raw term count alone would suggest. All Gaussian baselines are the published Pauli counts of Trenev et al. [7] (Table 5, Appendix B), cross-checked against analytic references (Szabo and Ostlund for H_2 ; analytical Slater-type orbitals for He).

c. Tier discipline. The scaling exponents are *measured* quantities (log-log fits with reported R^2), not derived; the angular selection rules underlying the ERI-density decay are *exact*; and the composed sparsity advantage *magnitude* is a measured quantity under a named ERI rule. The construction itself is zero-parameter: the Hamiltonian follows from nuclear charges and geometry alone.

IV. THE BENCHMARKED RESOURCE LIBRARY (PAPER 20)

Paper 20 turns the encoding into a usable artifact: the `hamiltonian()` registry of 37 systems (35 composed molecular qubit Hamiltonians spanning $Z = 1$ –56 (H through Ba), plus the He and H_2 reference atoms), including multi-center systems (CO , N_2 , F_2), mixed frozen-core molecules (NaCl), and the first transition series as hydrides (ScH through ZnH), exported to Qiskit, OpenFermion, and PennyLane.

a. Two architectures. The library exposes two composed-molecular architectures with complementary strengths. The *composed* (pseudopotential, PK) architecture is the sparser of the two: LiH requires 334 Pauli terms at 30 qubits, with $13\times$ fewer QWC groups than the Gaussian STO-3G reference at 12 qubits (raw baseline, pair-diagonal rule — the same conventions as the Pauli comparison next). The small-system market-test comparison “334 versus 907 STO-3G Pauli terms” is near parity once conventions are matched: STO-3G’s qubit-reduced count is 276 (vs. the composed 334), and

under the exact global- M_L rule LiH re-prices to 838 (0.92 $\times$ the raw 907). The broad “one-to-three-orders-of-magnitude sparser” advantage therefore rests on the scaling gap and the multi-center block structure, *not* on the small-molecule minimal-basis count—which is convention-sensitive and near parity. The *balanced coupled* (PK-free) architecture trades sparsity for chemistry: it binds LiH with an FCI minimum at a computed $R_{\text{eq}} = 3.227$ bohr (7.0% above the experimental 3.015 bohr) at $n_{\text{max}} = 2$ (878 Pauli terms, 30 qubits); a single-point evaluation at the experimental geometry reaches a 0.20% energy error at $n_{\text{max}} = 3$ when solved exactly (number-projected FCI).

b. Row-conditional chemistry scope. The chemistry-accuracy scope at $n_{\text{max}} = 2$ is honestly *row-conditional*: first-row hydrides (LiH, BeH₂, H₂O, ...) bind with documented PES topology, while second-row hydrides (NaH and below) exhibit monotone overattraction under every tested configuration—frozen-core balanced, multi-zeta valence, screened cross-center, explicit-core Hartree-Fock—with no interior PES minimum at the qubit-Hamiltonian level. This scope boundary is documented empirically, not papered over: the framework’s value proposition is structural sparsity with quantitative truncation-convergence rates (Paper 38 state-space GH convergence), not energy-accuracy parity with CCSD(T) on second-row systems.

c. Frozen cores. Second- and third-row molecules use analytical frozen cores ([Ne], [Ar], [Ar]3d¹⁰); the frozen-core energy enters only through the identity operator and does not affect the quantum measurement cost. Isostructural molecules with frozen cores produce identical Pauli counts across rows—the periodic-table structure is visible directly in the resource counts.

V. THE REPRESENTATION-THEORETIC FOUNDATION (PAPER 16)

The composed builder assembles molecules from atomic building blocks, and Paper 16 supplies the group-theoretic classification of those blocks. After separation of the hyperradius, the N -electron angular dynamics live on S^{3N-1} and the angular kinetic energy is the $\text{SO}(3N)$ Casimir operator with eigenvalue $\mu_{\text{free}} = \nu(\nu + 3N - 2)/2$. Fermionic antisymmetry restricts the spatial wavefunction to the S_N irreducible representation conjugate to the spin Young diagram, whose first-column length λ_1 fixes the minimum angular quantum number $\nu = N - \lambda_1$. For every atom with ground-state spin $S < N/2$ the spatial irrep has $\lambda_1 = 2$, giving the universal result

$$\nu = N - 2, \quad \mu_{\text{free}} = 2(N - 2)(N - 1). \quad (2)$$

This produces a five-type classification of atoms—A (trivial), B (noble gas), C (alkali), D (alkaline earth), E (open shell)—and recasts the periodic law as the repetition of the S_N irrep sequence $C \rightarrow D \rightarrow E \rightarrow B$ within each period. The classification is implemented

in `atomic_classifier.py`, which refines the open-shell type E into a p -block dispatch (E) and a separate d -block dispatch (F) for the composed builder, and supplies the universal angular quantum number $\nu = N - 2$.

a. Honest framing. The classification *maps onto*, rather than *predicts*, the known periodic table: its value is a unified group-theoretic language for a pattern usually described empirically, not a derivation of chemistry from first principles. Paper 16 also establishes a structural fact relevant to the relativistic extension: the topology (ν, μ_{free}) is smooth through $Z = 1/\alpha \approx 137$, so the Dirac instability is a *metric* singularity on S^{3N-1} , not a topological one.

VI. THE NUCLEAR EXTENSION AND ITS BOUNDARY (PAPER 23)

Paper 23 applies the angular-sparsity machinery to a system the conformal S^3 projection cannot reach—the nuclear shell model—and in doing so delineates exactly which part of the framework is universal.

a. What the angular machinery delivers. Built on the harmonic-oscillator single-particle basis with a Mayer-Jensen spin-orbit term, the construction reproduces the HO shell closures 2, 8, 20, 40, 70, 112 directly from graph state counting, and recovers the physical magic numbers 2, 8, 20, 28, 50, 82, 126 with a spin-orbit strength $v_{ls}/\hbar\omega \approx 0.17$ and a Nilsson $\ell(\ell + 1)$ correction $d_{\ell\ell}/\hbar\omega \approx 0.02$. The qubit encodings inherit the angular sparsity: the deuteron ($1p + 1n$) requires 16 qubits and 688 non-identity Pauli terms using the Minnesota nucleon-nucleon potential and Moshinsky-Talmi brackets; helium-4 ($2p + 2n$) requires 16 qubits and 828 Pauli terms despite a Hilbert space 12.25 $\times$ larger—only 1.20 $\times$ more terms, a direct consequence of the angular selection rules carrying through to multi-particle systems. A composed nuclear-electronic deuterium encoding (26 qubits) validates the multi-block architecture and quantifies the $\sim 10^{13}$ coefficient ratio between nuclear, electronic, and hyperfine scales.

b. The rigidity boundary.

Theorem 1 (Fock projection rigidity). *The S^3 conformal projection that makes electronic Hamiltonians equivalent to a graph Laplacian is unique to the Coulomb potential and does not extend to the harmonic-oscillator or Woods-Saxon systems.*

For nuclear Hamiltonians the GeoVac contribution is therefore the angular machinery and qubit encoding only, *not* the conformal projection. This is the structural dual of the foundations-side HO rigidity theorem (Paper 24).

c. Honest framing. All nuclear physics inputs (Minnesota parameters, spin-orbit strength, $\hbar\omega$) follow the standard shell-model literature—they are not GeoVac-derived—and the quantitative binding energies at $N_{\text{shells}} = 2$ are far from experiment. They are to be read as *encoding-validation benchmarks*, demonstrating

that the two-species sparse qubit architecture works, not as a nuclear-structure calculation.

VII. HONEST SCOPE AND POSITIONING

a. What is robust. The scaling laws are the load-bearing results and are robust: atomic $O(Q^{3.15})$, composed $O(Q^{2.5})$ universal, 1-norm $O(Q^{1.69})$, QWC $O(Q^{3.36})$, and the exact linearity $N_{\text{Pauli}} = cQ$ (1). These are unaffected by the ERI-rule choice (constant prefactor) and rest on the potential-independent angular-sparsity theorem (Paper 22). The qubit library is a concrete, exported, reproducible artifact.

b. What is conditional. The *absolute* sparsity multipliers and small-system market-test lines depend on the disclosed pair-diagonal ERI rule and re-price by a constant factor under the exact global- M_L rule (LiH to near parity with STO-3G). All Pauli comparisons are at matched qubit count, not matched accuracy: the composed basis carries 5–26% classical R_{eq} errors, the chemistry-accuracy scope is row-conditional (first-row binds; second-row overattracts), and the nuclear binding energies are encoding-validation benchmarks far from experiment.

c. What was tried and did not work. The arc includes several honest negatives that bound the design space. Discrete $\mathbb{Z}_2$ tapering beyond the Hopf $m_\ell \rightarrow -m_\ell$ symmetry does not transfer to relativistic chemistry: nei-

ther κ -parity, direct-basis m_j -parity, nor rotated-basis m_j -parity commutes with the Dirac-Coulomb Hamiltonian, because the jj -coupled Coulomb operator couples sign-counted sectors under sum-conservation (not parity) rules. Adding off-diagonal cross-block h_1 coupling to gain binding produces $16\times$ over-binding on LiH; the framework’s binding wall — the structural obstruction to chemistry-accuracy binding documented in the Group 2 record — is structurally inseparable from its encoding advantages. A non-abelian gauge upgrade of the composed Dirac does not reduce the Pauli count below the $\mathbb{Z}_2$ -tapered identity, because tapering requires a Hamiltonian *symmetry* and the gauge freedom is a basis change, not a symmetry. The spectral action is the wrong functional for chemistry observables at finite cutoff. These negatives are recorded so they are not re-derived.

d. Positioning. The natural target is variational, NISQ-era simulation—where the Pauli term count and the number of QWC measurement groups dominate cost—at fixed molecular geometry. The sub-quadratic Pauli 1-norm makes the encodings relevant to qubitization as well, but the framework is not positioned as a fault-tolerant-QPE or geometry-optimisation tool, and it is not a replacement for production quantum chemistry. It is a structurally sparse, zero-parameter, ecosystem-exported library of qubit Hamiltonians with quantitative truncation-convergence rates—a research instrument for the resource-estimation and algorithm-development side of quantum simulation.

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 - [2] J. Loutey, “Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory,” GeoVac Paper 14 (2026).
 - [3] J. Loutey, “The Periodic Table Recast as S_N Representation Theory on S^{3N-1} ,” GeoVac Paper 16 (2026).
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